

Ali Hamed

Embedded Software Engineer | C/C++, Linux, Real-Time Systems, Networking

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Languages: Arabic & Hebrew (native) • English (professional)

SUMMARY

Embedded-focused software engineer with **2 years** of industry experience developing C/C++ tests and automation for low-level embedded components in complex, real-time distributed 4G/5G Open RAN systems. Works close to the hardware — Linux internals, low-level debugging (gdb, strace), BMC/IPMI provisioning, packet and protocol analysis, and hardware/software integration on a 20+ server lab. Strong C/C++ and Python, solid CS fundamentals (memory management, multi-threading, data structures), and a networking foundation across the L1–L3 stack. **NVIDIA Networking certified**. M.Sc. Information Systems & AI (GPA 93).

EXPERIENCE

Parallel Wireless — Software Automation Engineer	09.2024 – present • 2 years
- Developed C/C++ unit tests for low-level embedded components in real-time, multi-threaded systems. - Validated memory safety, race conditions, timing, and concurrency under load. - Debugged hardware/software interfaces with gdb, strace, journalctl, and dmesg. - Analyzed packets and protocols with Wireshark, PCAP analysis, iPerf, and tcpdump. - Brought up a 20+ server lab: BMC/IPMI, servers, switches, NICs, firewalls, RF equipment. - Ran RF simulations with spectrum analyzers and channel emulators against 4G/5G Open RAN. - Validated components across TCP/UDP/IP, Ethernet, VLANs, and L1–L3 networking protocols. - Tuned Linux configuration and services for low-latency, real-time workloads (Ubuntu, CentOS). - Built Python/Bash tooling for provisioning, log parsing, and system-level regression. - Integrated tests into Jenkins/Groovy CI/CD across 200–500 cases; 8h → 2h (~75%). - Delivered LLM-assisted log analysis to production, accelerating root-cause investigation. - Hands-on AI: deliver solutions from development to production with measurable impact.	

EDUCATION

M.Sc., Information Systems & AI • University of Haifa	GPA 93 • Expected 2027
Coursework in Big Data Systems, Cybersecurity, Software Engineering, Machine Learning, and Deep Learning, with applied projects on real data and distributed systems.	
B.Sc., Computer Science • Ruppin Academic Center	GPA 83 • 2022 – 2025
Specialization in AI & Data Science. Strong foundation in OOP, data structures, algorithms, operating systems, computer architecture, and software development (C/C++, Python, Java).	

CERTIFICATIONS

NVIDIA Networking • Apr 2025	Certificate
NVIDIA AI Infrastructure and Operations Fundamentals • Sep 2025	Certificate
AWS Cloud Developing • Dec 2024	Certificate
AWS Cloud Practitioner Essentials • Nov 2024	Certificate
IBM Generative AI and LLMs • Apr 2025	Certificate

SKILLS

Languages	C • C++ • Python • Bash • Shell • Java • SQL • Groovy
Embedded & Systems	embedded systems • real-time systems • multi-threading • memory management • concurrency • Linux internals (Ubuntu, CentOS) • system services • low-latency tuning • board bring-up • firmware validation • hardware debugging • test benches • communication protocols
Low-Level Debugging	gdb • strace • journalctl • dmesg • Wireshark • PCAP analysis • iPerf • tcpdump • packet & protocol analysis
Networking	TCP/UDP/IP • Ethernet • VLANs • L1–L3 protocols • LTE/5G Open RAN • REST • WebSocket
Hardware & Lab	BMC/IPMI • servers • switches • RF equipment • hardware/software integration • KVM • VMware
Testing & CI/CD	C/C++ unit testing • PyTest • Robot Framework • Jenkins • Groovy • Git • Docker • Kubernetes
Foundations	data structures • algorithms • operating systems • computer architecture
RF & Telecom	spectrum analyzers • channel emulators • RF simulation & signal analysis • RF equipment • 4G/LTE • 5G NR • Open RAN • networking protocols • lab equipment (servers, switches, NICs, firewalls)
Applied AI Delivery	end-to-end AI delivery (prototype → production) • LLM agents • RAG • FAISS / vector search • LangChain • MCP / tool calling • prompt engineering • model serving (REST/FastAPI) • containerized deployment (Docker, Kubernetes) • GPU/CUDA • evaluation & error analysis • human-in-the-loop
Practices & Methodologies	Agile • SDLC • hardware/software integration • low-level debugging • state machines • event-driven programming • code review • version control • unit testing • attention to detail • problem-solving • technical documentation

VOLUNTEER

Volunteer Instructor • Hezkek LaAtid "Scientific Excellence", Fureidis High School	09.2025 – present
Teaching Computer Science fundamentals and programming to 40 students.	
Student Tutor • Ruppin Academic Center (Algebra, Python, Discrete Math)	01.2023 – 02.2024
"Perach" Program • Tutor & mentor for elementary school students	2022 – 2025